

An update on bioactive plant lignans.



Lignans are a class of secondary plant metabolites produced by oxidative dimerization of two phenylpropanoid units. Although their molecular backbone consists only of two phenylpropane (C6-C3) units, lignans show an enormous structural diversity. There is a growing interest in lignans and their synthetic derivatives due to applications in cancer chemotherapy and various other pharmacological effects. This review deals with lignans possessing anticancer, antioxidant, antimicrobial, anti-inflammatory and immunosuppressive activities, and comprises the data reported in more than 100 peer-reviewed articles, so as to highlight the recently reported bioactive lignans that could be a first step towards the development of potential new therapeutic agents.